

- 4 A battery B, a variable resistor R and a uniform resistance wire PQ are connected in series, as shown in Fig. 4.1

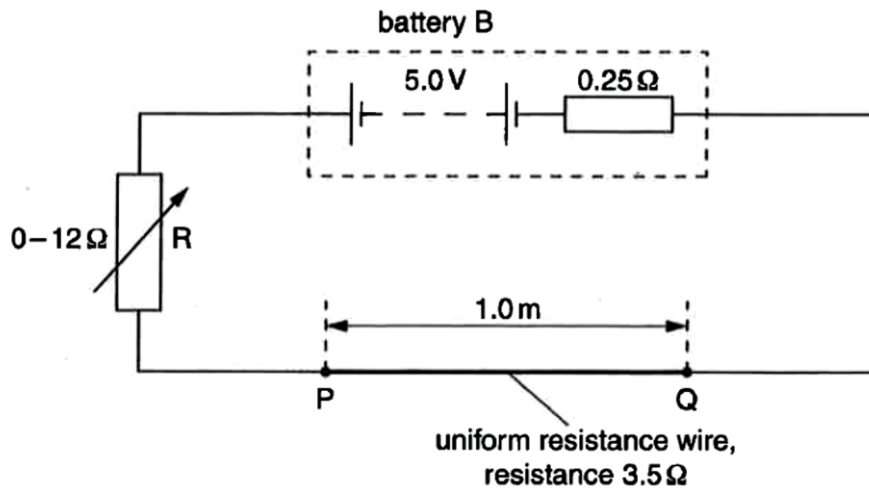


Fig. 4.1

Battery B has electromotive force (e.m.f.) 5.0 V and internal resistance 0.25 Ω.

Wire PQ has length 1.0 m and resistance 3.5 Ω at room temperature.

- (a) The resistance of R is set to 4.0 Ω.

Calculate, when the circuit is just turned on,

- (i) the potential difference across wire PQ,

p.d. = V [2]

Solution

$$\text{Current in circuit} = 5.0 / (0.25 + 4.0 + 3.5) = 0.645 \text{ A}$$

$$\text{the p.d. across wire PQ} = 0.645 (3.5) = 2.26 \text{ V}$$

OR,

$$\text{p.d. across wire PQ} = 5.0 - (0.645)(0.25 + 4.0) = 2.26 \text{ V}$$

OR,

$$\text{p.d. across wire PQ} = (5.0)(3.5) / (4.0 + 3.5 + 0.25) = 2.26 \text{ V}$$

M1
A1

- (ii) the percentage of total power transferred to wire PQ.

percentage = % [2]

[Turn over

		Solution Power transferred to wire $PQ = IV = 0.645(2.26)$ Power generated by Battery B $= IE = 0.645 (5.0)$ Percentage power transferred $= [\text{Power transferred to wire } PQ / \text{power generated by Battery B}] \times 100\%$ $= [0.645 (2.26) / 0.645 (5.0)] \times 100\%$ $= 45.2 \%$	M1 A1
	(b)	The temperature of the wire gradually increased from room temperature to maximum steady temperature.	
	(i)	Describe and explain the variation in the terminal potential difference (p.d.) across B. Numerical values are not required.	
			
			[2]
		Solution: As the temperature increases, its resistance increases, hence the total resistance increases and the current decreases terminal p.d. $= E - Ir$ will be increased since I decreases and E and r are constants	B1 B1
	(ii)	Suggest why the temperature of the wire will reach a steady maximum value.	
			
			[2]
		Solution: When the maximum temperature is reached, the electrical power absorbed is equal to the power dissipated to the surroundings. There is no net energy absorbed by the wire so the temperature will remain constant.	B1 B1

[Total: 8]

